

Database Technologies – Assignment 1

1. Create 'spj' database'. (Database for suppliers, parts and jobs)
2. use spj database;
3. Create following tables in spj database.

❖ CREATE TABLE S (S# CHAR(5) NOT NULL, Sname CHAR(20), status SMALLINT, city CHAR(15));

```
Field Type Null Key Default Extra
S# char(5) NO NULL
Sname char(20) YES NULL
status smallint(6) YES NULL
city char(15) YES NULL
4 rows in set (0.102 sec)

KD1_Rosh_98811> INSERT INTO S VALUES('S1', 'Smith', 20, 'London');
Query OK, 1 row affected (0.036 sec)

KD1_Rosh_98811> INSERT INTO S VALUES('S2', 'Jones', 10, 'Paris');
Query OK, 1 row affected (0.027 sec)

KD1_Rosh_98811> INSERT INTO S VALUES('S3', 'Blake', 30, 'Paris');
Query OK, 1 row affected (0.032 sec)

KD1_Rosh_98811> INSERT INTO S VALUES('S4', 'Clark', 20, 'London');
Query OK, 1 row affected (0.034 sec)

KD1_Rosh_98811> INSERT INTO S VALUES('S5', 'Adams', 30, 'Athens');
Query OK, 1 row affected (0.034 sec)

KD1_Rosh_98811> SELECT * FROM S;
+----+-----+-----+-----+
| S# | Sname | status | city |
+----+-----+-----+-----+
| S1 | Smith | 20 | London |
| S2 | Jones | 10 | Paris |
| S3 | Blake | 30 | Paris |
| S4 | Clark | 20 | London |
| S5 | Adams | 30 | Athens |
+----+-----+-----+-----+
5 rows in set (0.005 sec)

KD1_Rosh_98811> |
```

❖ CREATE TABLE P (P# CHAR(6) NOT NULL, Pname CHAR(20), color CHAR(6), weight SMALLINT, city CHAR(15));

```
KD1_Rosh_98811> SHOW TABLES;
+-----+
| Tables_in_spj |
+-----+
| P |
| S |
+-----+
2 rows in set (0.011 sec)

KD1_Rosh_98811> INSERT INTO P VALUES('P1', 'Nut', 'Red', 12, 'London');
Query OK, 1 row affected (0.058 sec)

KD1_Rosh_98811> INSERT INTO P VALUES('P2', 'Bolt', 'Green', 17, 'Paris');
Query OK, 1 row affected (0.030 sec)

KD1_Rosh_98811> INSERT INTO P VALUES('P3', 'Screw', 'Blue', 17, 'Rome');
Query OK, 1 row affected (0.025 sec)

KD1_Rosh_98811> INSERT INTO P VALUES('P4', 'Screw', 'Red', 14, 'London');
Query OK, 1 row affected (0.031 sec)

KD1_Rosh_98811> INSERT INTO P VALUES('P5', 'Cam', 'Blue', 12, 'Paris');
Query OK, 1 row affected (0.030 sec)

KD1_Rosh_98811> INSERT INTO P VALUES('P6', 'Cog', 'Red', 19, 'London');
Query OK, 1 row affected (0.044 sec)

KD1_Rosh_98811> SELECT * FROM P;
+----+-----+-----+-----+-----+
| P# | Pname | color | weight | city |
+----+-----+-----+-----+-----+
| P1 | Nut | Red | 12 | London |
| P2 | Bolt | Green | 17 | Paris |
| P3 | Screw | Blue | 17 | Rome |
| P4 | Screw | Red | 14 | London |
| P5 | Cam | Blue | 12 | Paris |
| P6 | Cog | Red | 19 | London |
+----+-----+-----+-----+-----+
6 rows in set (0.017 sec)

KD1_Rosh_98811> SELECT * FROM P;|
```

❖ CREATE TABLE J (J# CHAR(4) NOT NULL, Jname CHAR(10), City CHAR(15));

```
Query OK, 1 row affected (0.030 sec)
KD1_Rosh_98811> INSERT INTO J VALUES('J2', 'Punch', 'Rome');
Query OK, 1 row affected (0.034 sec)
KD1_Rosh_98811> INSERT INTO J VALUES('J3', 'Reader', 'Athens');
Query OK, 1 row affected (0.030 sec)
KD1_Rosh_98811> INSERT INTO J VALUES('J4', 'Console', 'Athens');
Query OK, 1 row affected (0.026 sec)
KD1_Rosh_98811> INSERT INTO J VALUES('J5', 'Collator', 'London');
Query OK, 1 row affected (0.030 sec)
KD1_Rosh_98811> INSERT INTO J VALUES('J6', 'Terminal', 'Oslo');
Query OK, 1 row affected (0.038 sec)
KD1_Rosh_98811> INSERT INTO J VALUES('J7', 'Tape', 'London');
Query OK, 1 row affected (0.032 sec)
KD1_Rosh_98811> SELECT * FROM J;
+-----+-----+-----+
| J# | Jname | City |
+-----+-----+-----+
| J1 | Sorter | Paris |
| J2 | Punch | Rome |
| J3 | Reader | Athens |
| J4 | Console | Athens |
| J5 | Collator | London |
| J6 | Terminal | Oslo |
| J7 | Tape | London |
+-----+-----+-----+
7 rows in set (0.005 sec)
KD1_Rosh_98811> |
```

❖ CREATE TABLE sp (S# CHAR(4) NOT NULL, P# CHAR(4) NOT NULL, J# CHAR(4) NOT NULL, QTY INT);

```
KD1_Rosh_98811> INSERT INTO sp VALUES('S5', 'P4', 'J4', 800);
Query OK, 1 row affected (0.039 sec)
KD1_Rosh_98811> INSERT INTO sp VALUES('S5', 'P5', 'J4', 400);
Query OK, 1 row affected (0.031 sec)
KD1_Rosh_98811> INSERT INTO sp VALUES('S5', 'P6', 'J4', 500);
Query OK, 1 row affected (0.619 sec)
KD1_Rosh_98811> SELECT * FROM sp;
+-----+-----+-----+-----+
| S# | P# | J# | QTY |
+-----+-----+-----+-----+
| S1 | P1 | J1 | 200 |
| S1 | P1 | J4 | 700 |
| S2 | P3 | J1 | 400 |
| S2 | P3 | J2 | 200 |
| S2 | P3 | J3 | 200 |
| S2 | P3 | J4 | 500 |
| S2 | P3 | J5 | 600 |
| S2 | P3 | J6 | 400 |
| S2 | P3 | J7 | 800 |
| S2 | P5 | J2 | 100 |
| S3 | P3 | J1 | 200 |
| S3 | P4 | J2 | 500 |
| S4 | P6 | J3 | 300 |
| S4 | P6 | J7 | 300 |
| S5 | P2 | J2 | 200 |
| S5 | P2 | J4 | 100 |
| S5 | P5 | J5 | 500 |
| S5 | P5 | J7 | 100 |
| S5 | P6 | J2 | 200 |
| S5 | P1 | J4 | 100 |
| S5 | P3 | J4 | 200 |
| S5 | P4 | J4 | 800 |
| S5 | P5 | J4 | 400 |
| S5 | P6 | J4 | 500 |
+-----+-----+-----+-----+
24 rows in set (0.005 sec)
KD1_Rosh_98811> |
```

1. How many orders data is present in the database?

Ans : 10 records are stored in orders table.

2. How many customers are present in the database?

Ans: 7 Customers are present in database.

3. How many salespeople are present in the database?

Ans: 5 Peoples are present in database.

4. On which date order with highest amount is placed?

Ans: On 1990-10-04 highest amount order is placed.

5. For which order salesman got maximum commission? Hint: you need to refer data of two tables.

Ans: on order number of 3001, 3006 salesman got maximum commission of 0.15%.

6. Which salesmen are handling more than two customers? Hint: you need to refer data of two tables.

Ans: No Salesman handling more than two customers every customer handling one or two customers only.

7. Which city have single customer?

Ans: Berlin city have only one customer.

8. Which city has multiple salespeople?

Ans: London city has multiple salespeople.

9. What is name of customer and salesman of the maximum amount order?

Ans: Customer Cisneros and salesman Peel has maximum amount order.

10. Study relationship between the tables in sales database.

Ans : orders table is related to Customer table and salespeople by cnum (customer number)

and snum (salesman number)

orders - cnum and snum are foreign key.

Customer – snum is foreign key.

Order.cnum = customer.cnum

Order.snum = salespeople.snum